Beneath the Surface: Which Screens See Which Cartels*

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Abstract

Between 2009 and 2019, Brazil's antitrust authority convicted seven procurement cartels operating on the electronic platform of the state of São Paulo (BEC), involving 35 firms and two distinct coordination technologies: five bid-rigging arrangements and one large hub-and-spoke scheme. We link these convictions to 2.6 million competitive auctions on BEC and to the universe of matched employer–employee records (RAIS), and report the first AUC-based evaluation of canonical detection screens against criminal ground truth in a middle-income setting.

On 2,319 close-margin auctions involving at least one convicted firm, classical variance screens yield AUCs of 0.46–0.56 in the aggregate. Disaggregating by coordination technology resolves the null into two very different findings. On the five bid-rigging cartels ($N = 342$), the coefficient-of-variation AUC is 0.65; on the hub-and-spoke cartel that accounts for 85% of the sample, it is at chance. The bid-rigging result survives cartel-level resampling (cluster-bootstrap 95% CI $[0.61, 0.75]$) and a label-permutation test (Bonferroni-adjusted $p = 0.002$). The close-margin regression discontinuity of Kawai et al. (2023) identifies a 27 percentage-point baseline incumbency jump on the full pregão sample but produces a null on the cartel-active subsample; we interpret this as cartels *disrupting* competitive path dependence rather than amplifying it.

A worker-flow screen constructed from the full RAIS panel appears to close the hub-and-spoke gap, with an AUC of 0.75 on the cartel-active sample. Temporal conditioning overturns this finding. Restricting worker flows to connections observable before the auc-

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tion year drops the AUC to 0.50. A temporal decomposition assigns the entire signal to post-conduct labor reallocation: workers flowing from convicted firms to their rivals after conduct ends, at a ratio rising from 1.4:1 to 4:1 over the panel. An exit placebo built on non-cartel BEC firms that stop bidding in the same period reproduces both the outflow magnitude (+1.6 pp vs. +1.9 pp for cartel firms) and the destination-concentration structure (34.4% vs. 14.5% going to product-market peers), establishing that the post-event reallocation is a signature of thin-market firm contraction rather than of cartel enforcement specifically.

Two implications follow. Classical variance screens work—but only for the coordination technology they were designed for; matching the screen to the suspected cartel architecture is essential, and hub-and-spoke arrangements require investigative instruments other than bid-pattern analysis. And network screens evaluated on panel administrative data are vulnerable to look-ahead bias whenever firm contraction—driven by enforcement or otherwise—triggers reallocation that the full panel absorbs; temporal conditioning is necessary before any panel-based network screen can be recommended to competition authorities.

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1. Introduction

Procurement cartels leave different statistical footprints depending on how they coordinate. Bid-rotation agreements compress the bid distribution around a designated winner, and classical variance screens were built to detect that compression. Hub-and-spoke cartels pre-arrange prices through a central intermediary and leave individual bid distributions approximately competitive, even when contract-level prices are fixed. A variance screen trained on rotation signatures should work in the first case and fail in the second. Whether that textbook logic holds in a conviction-anchored benchmark—and what happens when a new detection frame, built on external employment data, appears to close the hub-and-spoke gap—is the question of this paper.

We link seven procurement cartels convicted by Brazil's competition authority (CADE) between 2009 and 2019 to 2.6 million competitive auctions on the electronic platform of the state of São Paulo (BEC) and to the universe of formal employment relationships recorded in RAIS. The resulting benchmark is, to our knowledge, the first AUC-based evaluation of canonical procurement-detection screens against criminal ground truth outside a high-income setting. On the 2,319 close-margin pregão auctions in which at least one convicted firm bid during its conduct period, classical variance screens yield AUCs of 0.46–0.56 in the aggregate. Disaggregating by coordination technology changes the picture. On the five bid-rigging cartels ($N = 342$) the coefficient-of-variation AUC is 0.65; on the hub-and-spoke cartel that accounts for 85% of the sample, it is at chance. The bid-rigging result is robust to cartel-level resampling (cluster-bootstrap 95% CI $[0.61, 0.75]$) and to a label-permutation test that sidesteps the size of the clean-control sample (Bonferroni-adjusted $p = 0.002$). A close-margin regression discontinuity on firm-level incumbency delivers a 27 percentage-point baseline jump on the full pregão sample ($p < 0.001$) but a null on cartel-active auctions, which we read as cartels disrupting the repeat-winner pattern that bid-rotation agreements are designed to break.

We then turn to network-based detection. A worker-flow screen built from the full RAIS panel—counting employees who have worked at both the winning and runner-up firms—yields an AUC of 0.75 on the cartel-active sample, enough to close the hub-and-spoke gap at first pass. Temporal conditioning overturns this finding. Restricting worker flows to connections observable before the auction year brings the AUC down to 0.50, and a temporal decomposition locates the entire signal in post-conduct labor reallocation: workers flow from convicted firms to their rivals at a ratio rising from 1.4:1

to 4:1 over the panel. A placebo built on non-cartel BEC firms that stop bidding in the same period reproduces both the outflow magnitude (+1.6 pp vs. +1.9 pp for cartel firms) and the destination-concentration structure (34.4% vs. 14.5% of flows going to product-market peers). What the worker-flow screen captures is not collusion operating but firm contraction afterward; the screen absorbs the aftermath and overstates its real-time usefulness.

A growing literature uses statistical regularities in bidding behavior to infer collusion that is never documented directly. Classical screens exploit the compression that bid-rotation agreements impose on the bid distribution: prearranged prices cluster around the designated winner, yielding low dispersion, few active bidders, and predictable rotation (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). The complementary approach of Kawai et al. (2023) focuses on the winning–losing threshold: under genuine competition, firms that barely win and barely lose should look alike, and any sharp discontinuity at the cutoff identifies a non-competitive wedge. Both families of tests have been validated primarily against cartels in high-income economies with deep bidder pools and strong institutional capacity for enforcement; their track record in thinner settings rests on a small and growing empirical base.

Network-based methods have pushed the frontier in two directions. Within the procurement data itself, co-bidding networks flag candidate cartels through group-level cohesion and exclusivity (Wachs and Kertész, 2019; Signor et al., 2020; Huber and Imhof, 2019; Clark et al., 2018). Outside the bid record, a recent wave exploits external network layers: Asai and Charoenwong (2026) show that common-ownership links predict bid rigging in Singapore, and Villamil et al. (2024) build multiplex ownership–co-bidding networks for Swedish procurement. What none of these studies does—and what our benchmark uniquely enables—is test whether the detection signal from a network layer survives temporal conditioning: restricting the network measure to information available *before* the auction being screened. This distinction matters because network panels built from administrative registries (employment, ownership, board membership) mechanically include post-event observations, and any correlation between the network measure and the ground-truth label may reflect the consequences of detection and enforcement rather than the coordination the screen is meant to identify.

Our primary evaluation sample is a *cartel-active* benchmark of 2,319 close-margin pregão auctions in which at least one convicted firm bid during the active period of

its cartel (details of sample construction in Section 5). On that benchmark, classical screens disappoint. Bid dispersion and the coefficient of variation are weakly compressed at cartel-tainted auctions (Cohen's d between -0.07 and -0.24), and Welch t-statistics are significant thanks to the enormous control sample—but the aggregate AUCs top out at 0.56, far short of what a triage tool would require (though, as Section 6.2 shows, disaggregating by coordination technology recovers operationally useful detection for bid rigging). The close-margin RD produces a null. In thin markets—where the median number of bidders falls below five—classical screens actually fire in the *wrong* direction (AUC 0.40–0.45), consistent with phantom-bidding mechanisms that inflate rather than compress the bid distribution.

Worker-flow screens initially appear to resolve this failure. Counting the employees who have worked at both the winning and the runner-up firm over the 2009–2017 RAIS panel yields an AUC of 0.75. But this headline rests on the full nine-year employment panel, which includes worker movements that occurred years *after* the auction being screened. When we condition on temporal precedence—counting only workers who had appeared at both firms *before* the auction year—the AUC falls to 0.52, indistinguishable from chance. A three-year rolling window produces an essentially identical result (0.54). The prospective detection power of worker-flow screens is, on this benchmark, zero.

The temporal decomposition reveals where the signal actually lives. We classify each shared worker by when the inter-firm connection was first observable, relative to the cartel's conduct period. Pre-conduct worker flows produce an AUC of exactly 0.50. During-conduct flows produce 0.62. Post-conduct flows—worker movements that occurred *after* the cartel period ended—produce 0.77, accounting for nearly all of the headline result. The flows are directional: workers move predominantly *from* convicted cartel firms *to* their market rivals, at a ratio that rises from 1.4:1 in 2010 to 4:1 by 2016–2017. This pattern is consistent with enforcement-driven labor reallocation: as investigations proceed and conduct periods end, cartel firms shed employees who are absorbed by competitors in the same thin labor markets.

The paper makes four contributions. *First*, a positive detection result. Classical variance screens detect bid-rigging cartels at operationally useful levels (AUC ≈ 0.65) but are blind to the hub-and-spoke cartel that dominates the sample. The aggregate AUC of 0.52–0.56 is a composition effect, not a failure of the screens. Competition authorities adopting variance-based triage tools should match the screen to the suspected coordination technology, and should expect hub-and-spoke arrangements that coordinate

prices outside the auction to remain invisible to bid-distribution methods.

Second, a methodological result with broad reach. Network screens constructed from administrative panel data—whether the panel is employment, ownership, or board interlocks—are vulnerable to look-ahead bias when evaluated against conviction-anchored ground truth. Post-enforcement changes in the network inflate the apparent AUC. Temporal conditioning—restricting the network measure to information observable before the auction being screened—is a prerequisite for any prospective detection claim. The bias has not, to our knowledge, been documented or tested in the detection literature; it applies directly to the recent wave of ownership-network (Asai and Charoenwong, 2026; Villamil et al., 2024) and leadership-interlock (Herrera-Caicedo et al., 2025) detection work.

Third, a substantive result on the labor-market aftermath of thin-market firm contraction. Post-conduct worker flows from cartel firms concentrate on product-market peers rather than arbitrary employers, at a concentration ratio of about 3.6 workers per market-peer firm to every worker per bystander firm. Szerman (2023) documented that debarred procurement firms lose nearly half their employees; we show where those employees go. A placebo built on non-cartel BEC exit firms reproduces the same concentration pattern, so we do not claim the pattern is cartel-specific. What we document is a feature of thin-market BEC-firm contraction that cartel enforcement triggers along with other causes.

Fourth, empirical infrastructure. We deliver the first AUC-based stress test of canonical procurement-detection screens against criminal ground truth outside a high-income setting. The benchmark—CADE Tribunal rulings linked to BEC bid-level records and RAIS employer–employee data at the firm level—is reusable and extends naturally to other Brazilian procurement platforms and to jurisdictions with comparable administrative-data layers.

The remainder of the paper proceeds as follows. Section 2 situates our work in the detection, network, and worker-mobility literatures. Section 3 describes public procurement in São Paulo and the CADE cartel cases that anchor the ground truth. Section 4 presents the empirical strategy—including the temporal-conditioning test that distinguishes prospective from retrospective signal—and a framework for screen heterogeneity across market structures. Section 5 describes the data sources and sample construction. Section 6 reports the results in two parts: the apparent detection result and its temporal unraveling. Section 7 discusses implications and concludes.

2. Related Literature

This paper draws on three literatures: the design and validation of cartel-detection screens, the use of network structure to identify collusion, and the economics of inter-firm worker mobility.

2.1. Detection through bid-pattern screens

The idea that bid-rotation agreements leave statistical footprints is now well established. Porter and Zona (1993) showed that collusive phases of the 1880s railroad freight cartel exhibit systematically lower bid dispersion than competitive phases, exploiting the cartel's own internal records as an exogenous regime switch. Pesendorfer (2000) documented bid rotation in Florida school-milk procurement; Bajari and Ye (2003) proposed exchangeability tests for bidder costs in Midwestern highway paving; Conley and Decarolis (2016) extended pair-level correlation tests to Italian procurement. Imhof et al. (2018) demonstrated that combining multiple screens in a portfolio outperforms any single indicator in Swiss data, and Abrantes-Metz et al. (2006) offered practical guidance on implementing variance screens while cautioning against single-indicator reliance. Harrington (2008) synthesized the design principles; the informativeness of any screen depends on the competitive counterfactual and on the coordination technology the cartel actually uses, a point the heterogeneity results below make concrete.

A complementary strand focuses on the bid margin itself. Kawai et al. (2023) proposed a regression discontinuity centered on the winning–losing threshold in Japanese construction procurement: under genuine competition, narrowly winning and narrowly losing firms should look alike, and any discontinuity at the cutoff identifies a non-competitive wedge. Chassang and Ortner (2019) developed a theoretical framework in which bid constraints endogenously shape the collusive technology—predicting, among other things, that format-specific screens may outperform generic variance tests, a prediction consistent with the format heterogeneity we document below.

A common thread across these contributions is that nearly all validation exercises rely on ground truth from high-income economies with deep bidder pools and well-resourced enforcement agencies. Levenstein and Suslow (2006) documented across 81 international cartels that market structure, transparency, and enforcement capacity jointly determine cartel success—suggesting that screens validated in one institutional regime need not

travel well to another. Direct validation against criminal ground truth in a middle-income setting with structurally different labor markets remains scarce, and that is the gap our conviction-anchored benchmark is designed to fill.

2.2. Network-based and machine-learning approaches

A parallel literature has moved beyond univariate bid statistics to exploit the relational structure of procurement markets. Wachs and Kertész (2019) showed that co-bidding networks in Georgian procurement flag candidate cartels through group-level cohesion and exclusivity, even without criminal ground truth. Signor et al. (2020) built a probabilistic detector from the data released by Brazil's *Operação Lava Jato*, reporting 81–96 percent accuracy on the Petrobras construction cartel. Huber and Imhof (2019) combined classical screens with supervised machine-learning classifiers in Swiss data, correctly predicting 84 percent of collusive tenders out of sample. Clark et al. (2018) documented how collusive networks in Québec construction co-evolve with political connections, illustrating the interplay between bid rigging and corruption.

A more recent wave extends the network logic to data sources *outside* the bid record. Asai and Charoenwong (2026) link shareholder registries to procurement auctions in Singapore and show that common-ownership links predict identical bids; removing the ownership-network effect would cut government spending by 4–6 percent. Villamil et al. (2024) build temporal multiplex networks that combine ownership ties and co-bidding patterns in Swedish procurement, finding that denser ownership connections predict higher single-bidding and more missing bidders. Together, these papers establish that external network layers—corporate ownership, in their case—carry detection signal that co-bidding statistics alone cannot reach.

We add a third layer: labor-market connectivity, constructed from matched employer–employee records. Worker-flow networks differ from ownership networks in that they are available wherever governments maintain compulsory employer–employee registries (RAIS in Brazil, DADS in France, IAB in Germany), making the approach portable without new data collection. And because worker-flow data are informationally independent of both the bid record and the corporate registry, the risk of circularity between the screen and the behavior it targets is lower than for co-bidding or ownership-based approaches. Whether worker-flow connectivity captures *coordination* in real time or *enforcement aftermath* after the fact is, however, an open question—one that requires temporal conditioning to answer.

2.3. Worker mobility as an inter-firm channel

Labor economists have long recognized that workers carry knowledge between firms. Stoyanov and Zubanov (2012) showed that productivity gains spill across Danish firms through the mobility of their employees, with larger spillovers when the productivity gap between sender and receiver is wide. Serafinelli (2019) documented a similar pattern in Italy's Veneto region: workers arriving from high-productivity firms raise wages and output at their new employers. Jarosch et al. (2021) developed an equilibrium model in which team composition and worker reallocation jointly shape firm-level human capital. Worker flows are therefore not only a labor-market phenomenon but a conduit for inter-firm information transmission.

Two recent papers connect this conduit to collusion. Lamichhane and McGee (2024) provide experimental evidence that, in Bertrand markets, worker mobility between firms facilitates collusive pricing: workers carry pricing-strategy information, and firms that exchange employees coordinate on higher prices and more market sharing, especially in concentrated markets. Herrera-Caicedo et al. (2025) document that common leadership—two firms sharing executives or board directors—predicts collusion, using the Silicon Valley no-poach case evidence as a source of labor-market collusion ground truth: the probability of collusion between two firms rises by 11 percentage points after the onset of common leadership, a ninefold increase over the baseline rate. Their result is about labor-market collusion (wage-fixing and no-poach agreements) rather than product-market coordination, but it establishes the broader principle that shared inter-firm ties facilitate coordination through subtle channels. Our benchmark is designed to test the adjacent question: in the procurement-cartel setting, when leadership ties are absent, do worker-flow screens carry prospective detection signal—or do they merely absorb the labor-market aftermath of enforcement?

On the data-infrastructure side, two papers have already linked RAIS to Brazilian procurement—though not for detection. Barbosa and Straub (2020) track worker movements between public agencies and private suppliers, showing that supplier-to-administration revolving doors are detrimental to procurement efficiency. Szerman (2023) exploits a debarment-policy change to estimate the worker-level costs of anti-corruption enforcement, finding that blacklisted firms lose nearly half their employees. Both papers prove the RAIS–procurement linkage is feasible and productive; neither uses it as a cartel screen.

Taken together, Lamichhane and McGee (2024) establish the mechanism (worker mobility as a collusion channel), Asai and Charoenwong (2026) and Villamil et al. (2024) show that external network layers carry detection signal, and Barbosa and Straub (2020) and Szerman (2023) demonstrate the RAIS–procurement data infrastructure. None of the studies above conditions the network measure on temporal precedence: each constructs the screen from the full employment, ownership, or co-bidding panel, then evaluates it against convictions that occurred during or before the panel period. If enforcement changes the network—through labor reallocation, ownership restructuring, or altered bidding patterns—the screen absorbs the aftermath and overstates its real-time usefulness. This paper provides the first direct test of that distinction.

3. Institutional Background

3.1. Public procurement in São Paulo: the BEC platform

São Paulo's state government purchases common goods and services through a decentralized network of roughly 1,300 public buyer units (PBUs), all of which have been required since 2007 to transact on the state's electronic platform, the *Bolsa Eletrônica de Compras* (BEC). The scale is substantial: in 2018 alone, BEC handled approximately US$1.7 billion in procurement, and over the study period the platform processed more than R$7.4 billion for the State Health Secretariat and other agencies.

The BEC catalog organizes items by group and class—from pharmaceuticals to office supplies—totaling about 6,350 distinct product codes. Each tender, called a purchase offer (PO), may contain multiple items; each item is auctioned separately, so a unique procurement event is defined by a (PO $\times$ item code) pair, which we label POI. The platform records the fields a detection exercise requires: the reference price set by the buyer, every bid submitted, the identities of bidders and the winning firm, and the auction format.

Two formats predominate. In the *convite* (sealed-bid tender), firms submit confidential bids by a deadline; the lowest qualified bid wins. This format is typically used for purchases below R$176,000 (about US$35,000). In the *pregão eletrônico* (descending auction), firms first submit sealed proposals, after which an open phase begins: bidders observe the current lowest price in real time and can underbid for an initial 20 minutes, with automatic 4-minute extensions whenever a late bid arrives. The auction closes after

4 consecutive minutes of silence. The pregão carries no value ceiling and accounts for the majority of BEC spending.

The distinction matters for collusion. The pregão's iterative, anonymous bidding gives cartel members the ability to monitor compliance in real time but also exposes deviations; the convite's single-shot format makes monitoring harder but bid coordination simpler. Our primary analysis focuses on pregão auctions, for which the close-margin RD framework is most naturally motivated; convite results are reported as a complementary check.

3.2. Antitrust enforcement: CADE and the ground truth

Brazil's federal competition authority, the *Conselho Administrativo de Defesa Econômica* (CADE), is an autonomous agency within the Ministry of Justice. Its Tribunal renders administrative decisions—binding fines and behavioral remedies, not criminal sentences in the judicial sense. Cases typically begin with a complaint or a leniency application (*Termo de Compromisso de Cessação*, TCC), proceed through a formal investigation by the *Superintendência-Geral*, and conclude with a Tribunal vote.

Our ground truth draws on CADE proceedings in which the Tribunal convicted firms of cartel conduct in São Paulo procurement markets. We searched CADE's public repository for proceedings with final decisions that (i) name specific firms with identifiable CNPJs, (ii) specify the product market, and (iii) delineate a conduct period. Seven cartels meet these criteria, spanning sectors as varied as pharmaceuticals, school meals, trash bags, solar water heaters, and metropolitan rail, and involving 35 firms whose CNPJ roots we can match to BEC records.

The case files specify not only who colluded but when, and in which market. We use this granularity to label individual auctions as cartel-tainted, matching each convicted firm's CNPJ root, the CADE-specified conduct window, and the relevant product code against BEC tenders. The resulting benchmark captures both the identity and the timing dimensions of collusion, rather than treating every auction involving a flagged firm as equally suspect.

4. Empirical Strategy

We bring five detection approaches to the conviction-anchored benchmark: a close-margin regression discontinuity (Section 4.1), classical bid-distribution screens (Sec-

tion 4.2), network screens built from worker-flow and co-bidding data (Section 4.3), a specialization placebo (Section 4.6), and—crucially—a temporal-conditioning test that distinguishes prospective from retrospective detection power (Section 4.5). Every screen is evaluated by its area under the ROC curve (Section 4.4). A simple model in Section 4.7 organizes predictions about where classical screens should fail and where post-enforcement labor reallocation should concentrate.

4.1. Close-margin regression discontinuity

Following Kawai et al. (2023), we focus on auctions decided by a hair. Define the margin of victory in auction a as

$$MV_a = \frac{b_a^{\text{runner-up}} - b_a^{\text{win}}}{b_a^{\text{win}}}.$$

When MV_a is close to zero, the winner barely underbid the nearest rival. Under genuine competition, narrowly winning and narrowly losing firms should look alike on pre-determined characteristics; any discontinuity at the cutoff signals something non-competitive.

We estimate the jump using local-polynomial regression with a triangular kernel and the MSE-optimal bandwidth selector of Calonico et al. (2014), implemented via `rdrobust` (Calonico et al., 2017). Bandwidth values chosen for each specification are reported in the notes of the corresponding tables (typically $h \approx 0.015$–0.025 on our main samples). Two outcomes carry interpretive weight. *Incumbency*—whether the firm won the previous auction in the same market—tests for bid rotation: a jump would indicate that designated winners are systematically past winners. *Last bidder*—whether the firm submitted the final bid—tests for strategic timing. A density test on the running variable checks whether the margin itself is being manipulated.

4.2. Classical bid-distribution screens

We compute four screens on firm-best bids (one bid per firm per auction, keeping the lowest when multiple rounds exist):

1. **Coefficient of variation** (CV_a): standard deviation divided by mean. Bid rotation predicts lower CV because cover bids cluster near the designated winner (Porter and Zona, 1993; Abrantes-Metz et al., 2006).

2. **Bid spread** (S_a): $(\max - \min)/\text{mean}$. Like CV, compressed under cover bidding.
3. **Margin of victory** (MV_a): the designated winner should undercut cover bids by just enough to win, yielding a small margin.
4. **Number of bidders** (N_a): entry deterrence or market division should reduce the count (Marshall and Marx, 2012).

For each, we compare cartel-tainted auctions to clean controls on the close-margin benchmark ($|MV| \leq 0.10$), reporting the Welch t-statistic, Cohen's d, the Wilcoxon rank-sum p-value, and the AUC.

4.3. Network-based screens

We step outside the bid data and build three screens from administrative sources.

Worker-flow screens. For each (winner, runner-up) pair we measure labor connectivity from the full 2009–2017 RAIS panel:

- **Shared workers** (W_{ij}): the count of distinct PIS numbers appearing at both firms during the nine-year window.
- **Jaccard similarity** ($J_{ij} = W_{ij}/(N_i + N_j - W_{ij})$): normalizes the count by firm size, so that large employers are not mechanically flagged.

If cartel coordination in thin labor markets runs through personnel—managers carrying pricing information, technicians rotating between member firms—the employment records should show it (Stoyanov and Zubanov, 2012; Serafinelli, 2019). If instead the signal is retrospective, reflecting labor reallocation after the cartel period ends, it will appear in the full-panel measure but vanish under temporal conditioning. We test both possibilities.

Co-bidding screen. For each pair we count the auctions in which both firms submitted bids (C_{ij}). This provides a network channel built entirely within the procurement data.

4.4. Detection evaluation: AUC

We use the area under the ROC curve to compare screens. The AUC is simply the probability that a randomly chosen cartel auction scores more "cartel-like" than a randomly chosen clean auction. For screens where lower is more suspicious (CV, S, MV, N), we negate scores so that AUC > 0.5 always means detection in the predicted

direction. We compute the AUC via the Mann–Whitney U statistic with average-rank tie-handling:

$$\text{AUC} = \frac{U_1}{n_1 \times n_0},$$

where n_1 and n_0 are the cartel and clean sample sizes. All screens are evaluated on the same unified close-margin benchmark, so comparisons are apples-to-apples. Appendix A.1 shows that results are stable across bandwidths from 0.02 to 0.50.

4.5. Temporal conditioning and decomposition

Network screens built from administrative panel data face a fundamental identification challenge: the measure uses information from the *entire* panel, including periods after the auction being screened. If cartel detection or enforcement changes the network—for instance, by triggering labor reallocation from convicted firms to competitors—then the full-panel AUC conflates prospective detection power with retrospective correlation. We implement two tests.

Temporal conditioning. For each auction at year t, we recompute the worker-flow and co-bidding screens using only data from years strictly before t. If the AUC survives, the screen captures information available at the time of the auction. If it collapses, the signal is retrospective. We report a cumulative backward window $[2009, t-1]$ and a rolling window $[\max(2009, t-3), t-1]$. A limitation of this implementation is granularity: RAIS reports the calendar years in which a worker held a bond at a firm, not the months. Worker movements that occurred in January of year t are therefore not usable to screen an auction held in December of the same year, even though they are chronologically prior. Our temporal conditioning is thus conservative (it may discard some chronologically-prior information), not aggressive (it does not use chronologically-posterior information). A sub-annual implementation using job start and end dates is a natural refinement.

Temporal decomposition. For each shared worker between a cartel firm and a rival, we classify the inter-firm link by when it was first observable relative to the cartel's conduct period: **pre-conduct** (both firms employed the worker before the start year), **during-conduct** (the link appeared during $[\text{start}, \text{end}]$), or **post-conduct** (the link appeared after the end year). We then use each window's shared-worker count as a separate screen and evaluate it against the benchmark. This decomposition identifies *when* the signal arises—and whether it reflects coordination, ongoing cartel activity, or enforcement aftermath.

4.6. Specialization placebo

Elevated worker flows between a cartel firm and its rival could reflect several forces: pre-existing personnel connections, the natural circulation of a thin labor pool, or post-conduct labor reallocation. To separate market-level circulation from the cartel-specific component, we build a three-way comparison. The **cartel** sample consists of auction-level (winner, runner-up) pairs with at least one cartel-active firm (N = 2,319). The **placebo** consists of non-cartel pairs bidding on the same BEC item code in the same year. The **clean** baseline is every other non-cartel pair.

If the entire worker-flow signal were market specialization, the placebo should match the cartel sample. Any excess in the cartel sample over the placebo is a cartel-specific residual—though, as we show in the results, this residual is driven primarily by post-conduct displacement rather than by coordination during the cartel period.

4.7. A framework for screen heterogeneity

Screen effectiveness and the labor-market aftermath of enforcement both depend on market thickness. We sketch a minimal framework that organizes the predictions we take to the data. Consider $n \geq 2$ firms operating in a product market with a specialized-labor pool of thickness L.

Classical bid-distribution screens exploit the compression that bid rotation imposes on the bid distribution. In thick markets (large L, many bidders), rotation is the natural coordination device and the compression is detectable. In thin markets (small L, few bidders), the bid distribution is inherently noisy—few firms generate high variance even under collusion—and phantom-bidding or cover-bid mechanisms may inflate rather than compress dispersion. Classical screens should therefore work in thick markets and fail in thin ones.

The labor-market aftermath of enforcement has a parallel structure. When a cartel firm contracts—losing contracts, facing fines, undergoing investigation—it sheds workers. In thick labor markets, those workers disperse across many potential employers and the per-firm flow is diffuse. In thin labor markets, the same workers have few alternative employers, and the flows concentrate in the small set of firms that operate in the same specialized product market. The worker-flow screen, evaluated on the full panel, captures this concentration—but the concentration is one of *displacement*, not of *coordination*.

Four testable predictions follow:

1. **Coordination-technology gradient.** Classical screens exploit bid-distribution compression—a signature of bid rigging. Coordination that operates outside the auction (e.g., hub-and-spoke arrangements where a central intermediary sets prices upstream) does not compress the bid distribution and should be invisible to classical screens. Classical AUCs should therefore be high for bid rigging and near chance for hub-and-spoke cartels.
2. **Thickness gradient for classical screens.** In thin markets, classical AUCs should be near or below chance even for bid rigging, because few bidders generate high variance mechanically. In thick markets, they should recover. This prediction is independent of the temporal-conditioning critique and uses contemporaneous bid data.
3. **Thickness gradient for post-conduct reallocation.** Full-panel worker-flow AUCs should be highest in thin markets—not because coordination runs through personnel, but because displacement concentrates where the labor pool is shallowest.
4. **Temporal asymmetry.** If the worker-flow signal reflects coordination, it should be present in pre-conduct flows. If it reflects enforcement aftermath, it should be concentrated in post-conduct flows. The temporal decomposition directly tests this.

The framework is deliberately spare. It abstracts from repeated-game dynamics, leniency, and heterogeneity across auction formats. We offer it not as a structure to be estimated but as a lens for organizing the results and for specifying *when* each family of screens should and should not carry signal.

5. Data

The analysis rests on three administrative data sources—procurement bids, employment records, and antitrust rulings—which we link at the firm level to construct the conviction-anchored benchmark.

5.1. BEC: bid-level procurement data

We draw on the complete bid-level records from São Paulo's *Bolsa Eletrônica de Compras* (BEC) for January 2009 through December 2019. After restricting to pregão

and convite procedures with at least two participating firms, the data contain 2,613,010 competitive auctions, 13,957,419 firm-level bid participations, 40,273 distinct firms, and 153,048 unique item codes. Each record identifies the purchase offer, item code, participating firm (by CNPJ), bid amount, outcome, auction format, and purchasing agency.

For each auction we compute the screens described in Section 4.2—coefficient of variation, bid spread, margin of victory, and bidder count—on firm-best bids (keeping each firm's lowest bid when multiple rounds exist). Markets are defined transitively following Kawai et al. (2023): two firms share a market if they ever bid on the same item code in the same auction. This yields 7,581 distinct markets.

5.2. RAIS: employer–employee matched panel

The *Relação Anual de Informações Sociais* (RAIS) is Brazil's compulsory annual census of formal employment. Every establishment with a CNPJ must report each employment bond active at any point during the year, identified by the worker's PIS/PASEP number. We use the 2009–2017 vintages, restricting to establishments whose 8-digit CNPJ root appears in the BEC firm universe—roughly 19,000 roots. The result is a nine-year panel of every formal-sector job at every BEC-active firm.

From this panel we build the worker-flow edge list. For each pair of BEC firms (i, j), we count distinct PIS numbers that appear at both firms at any point during the nine years (the shared-worker count W_{ij}) and compute the Jaccard similarity J_{ij}. Pairs with no shared workers are zero-imputed when merged to the auction benchmark.

A temporal asymmetry is worth noting. The RAIS panel covers 2009–2017, while BEC auction data extend to 2019 and CADE judgment dates for most cartels fall after 2019 (Table 1). The "post-conduct" window we measure in the temporal decomposition (Section 6.8) therefore captures the period between the end of cartel conduct and the end of the RAIS panel—typically the investigation and pre-judgment period, not the post-judgment period. This distinction matters for interpreting enforcement-driven labor reallocation: the outflows we document may reflect the anticipation of enforcement rather than its formal realization.

5.3. CADE: cartel ground truth

Our ground truth comes from CADE administrative proceedings with final Tribunal decisions convicting named firms of cartel conduct in São Paulo procurement markets (see Section 3). Table 1 lists the seven cartels we identified—spanning pharmaceuticals,

school meals, trash bags, solar water heaters, metropolitan rail, airport cafeterias, and school transport—together with their conduct periods and the 35 firms whose CNPJ roots match BEC records.

The key feature for detection is that each case file specifies not just the colluding firms but the product market and time window. We exploit this to label individual auctions as cartel-tainted: an auction earns the flag only if a convicted firm bid during the CADE-specified conduct period for the relevant product and year. This firm × period × product matching avoids the coarser alternative of treating every auction involving a flagged firm as suspicious.

Table 1: Convicted Cartels in São Paulo Procurement Markets

Sector	CADE process	Period	Firms	In BEC	Conduct type
Trens e metrôs	08700.004617	1999–2013	13	10	Bid rigging
Merenda escolar	08012.010022	2006–2013	7	6	Market division
Medicamentos	08012.002222	2007–2011	8	8	Hub-and-spoke
Sacos de lixo	08700.005789	2008–2014	3	3	Bid rigging / price fixi
Aquecedores solares	08012.001273	2009–2010	5	4	Bid rigging
Cafeterias Infraero	08700.007278	2014	6	3	Bid rigging
Transporte escolar	08700.005876	2019	2	2	Bid rigging
Total (unique CNPJ roots, firms deduplicated)			35	—	
Total (cartel memberships)			—	36	

Notes: "Firms" and "In BEC" count unique 8-digit CNPJ roots per cartel. The two totals differ because one firm participates in two cartels: the unique-firm total is 35, the total count of cartel memberships is 36. Conduct periods hand-curated from CADE Tribunal decisions and press releases (see Appendix). Source: script 12.

5.4. Sample construction

All detection tests use the close-margin envelope $|MV| \leq 0.10$. Within that envelope we define three samples:

1. **Cartel-active benchmark** ($N = 2{,}319$ pregão). At least one convicted firm bid during the active period of its cartel. This is the primary evaluation sample. Of the seven CADE cartels listed in Table 1, five contribute ≥ 6 close-margin pregão auctions to the benchmark; *cafeteria aeroporto* contributes 1 and *aquecedores solares* contributes 0, reflecting their small scale and short conduct windows. The

bid-rigging subpool ($N = 342$) is therefore identified primarily from five cartels, a point we return to in the cartel-level robustness analysis (Section 6.3).

2. **Strict pair-matched sample** ($N = 10$ pregão). Both the winner and the runner-up belong to the same convicted cartel during the same conduct period. The gold standard for head-to-head coordination, but power-limited.
3. **Anytime-listed sensitivity** ($N \approx 80$ pregão). Both firms appear in a CADE conviction for the same cartel, but the auction year may fall outside the conduct window. Of the 107 pre-filtering auctions, 93 are out-of-window and 100 come from a single case (the medicamentos hub-and-spoke, CADE 08012.002222/2011-09). Reported as a sensitivity only.

The *clean control* is every close-margin auction with no cartel-listed firm ($N =$ 527,259 pregão; 519,557 convite). All screens are evaluated against this common baseline.

5.5. Variable definitions

Table 2 collects the key variables.

Table 2: Variable Definitions

Variable	Definition
MV_a	Margin of victory: $(b_a^{\text{runner-up}} - b_a^{\text{win}})/b_a^{\text{win}}$
CV_a	Coefficient of variation of firm-best bids
S_a	Bid spread: $(\max - \min)/\text{mean}$ of firm-best bids
N_a	Number of distinct firms submitting bids
Incumbent_a	$= 1$ if the winning firm won the preceding auction in the same market
Last bidder_a	$= 1$ if the winning firm submitted the final bid
W_{ij}	Shared workers: distinct PIS appearing at both firms i and j
J_{ij}	Jaccard similarity: $W_{ij}/(N_i + N_j - W_{ij})$
C_{ij}	Co-bidding count: auctions where both i and j bid

6. Results

The classical-screens result is organized around the coordination-technology heterogeneity that is the paper's headline finding. Section 6.1 establishes the aggregate baseline on the cartel-active pregão sample ($N = 2{,}319$), the close-margin RDD, and the

small-effect-size interpretation of the Welch t-statistics. Section 6.2 disaggregates by coordination technology and reports the AUC = 0.65 bid-rigging result and the at-chance hub-and-spoke result that drive the paper's lead claim. Sections 6.3 and 6.4 stress-test that claim with cartel-level resampling and an identification decomposition. Section 6.5 defends the competitive-path-dependence interpretation of the RDD baseline with a placebo battery and a pregão replication. Sections 6.6–6.8 turn to worker-flow screens and the temporal-conditioning critique; Sections 6.9–6.12 characterize the post-conduct labor reallocation pattern and compare it to a non-cartel exit placebo. Sections 6.13–6.15 reinterpret the market-thickness and specialization-placebo analyses in light of the temporal finding, and report per-cartel AUCs.

6.1. Classical screens and the close-margin RD

Table 17 presents the unified benchmark comparison of all screens on the same close-margin pregão sample ($|MV| \leq 0.10$). Panel A reports the classical bid-distribution screens on the cartel-active benchmark (N = 2,319 cartel-tainted auctions vs. N = 527,259 clean controls).

All four classical screens fire in the direction predicted by the bid-rotation model: the coefficient of variation ($\Delta = -0.035$, $d = -0.07$), bid spread ($\Delta = -0.173$, $d = -0.10$), and number of active bidders ($\Delta = -0.87$, $d = -0.24$) are compressed at cartel-tainted auctions relative to clean controls. The effect sizes are small: Cohen's d ranges from -0.07 to -0.24, indicating near-total distributional overlap between cartel and clean auctions. The AUCs reflect this overlap, ranging from 0.52 (coefficient of variation) to 0.56 (number of bidders)—above chance but far too low for investigative triage. Welch t-statistics are large (the large clean-control sample makes them so), but we treat Cohen's d and AUC as the primary effect-size measures throughout; the full-sample t-statistics reported in Table 17 should not be read as evidence of a large effect size. To sidestep the sample-size inflation in the t-tests entirely, we compute a permutation-test p-value on each AUC under the null of no difference between cartel-active and clean-control distributions (B = 2,000 permutations, seed 20260418). The permutation p-values for the CV, bid-spread, and bidder-count screens are below 0.001 on the full cartel-active sample (N = 2,319) and remain so after a Bonferroni correction for four simultaneous tests (adjusted $p = 0.002$); on the bid-rigging subpool the same three screens are significant at the same level, while the margin-of-victory screen fails the permutation test in both samples ($p_{\text{perm}} = 0.48$ on bid-rigging, $p_{\text{perm}} = 1.00$ on the

full sample), consistent with its wrong-direction point estimate.

The margin of victory exhibits an anomalous pattern: $\Delta MV = +0.004$ ($d = +0.17$), with an AUC of 0.46—slightly *below* chance and in the *opposite* direction to the bid-rotation prediction. Cartel-active auctions have marginally *larger* winning margins, not smaller. We interpret this as consistent with a partial phantom-bidding mechanism in which cover bids are inflated relative to the designated winner's price, widening rather than compressing the margin.

A direct replication of the Kawai et al. (2023) close-margin RD detects a baseline incumbency advantage on the full sample of 261,370 close-margin convite observations at bandwidth 0.02 ($\hat{\beta} = 0.018$, SE $= 0.005$, $p < 0.001$), using the firm-specific outcome *won the last auction the firm participated in within the same market* (`won_t_minus_1_market`). This is the design reported in the project's earlier working version. It reflects a competitive pattern in which recent winners are slightly more likely to win again. On the cartel-active subsample ($N \approx 488$ convite firm-auctions), no *additional* discontinuity is detected ($p > 0.15$); the sample size is small, and the minimum detectable effect at 80% power is roughly 0.57—so a cartel-specific jump of the magnitude of the baseline (+1.8 pp) is about 30× below detectability. A density test on the close-margin running variable detects no manipulation in either the pregão or the convite sample ($T_{\text{CJM}} \approx -0.02$, $p > 0.98$ in both). We return to the interpretation of the baseline and its robustness in Section 6.5.

Sample note. The baseline RDD just summarized is estimated on convite, the platform format for contracts below roughly US$35,000. All remaining results in this paper—classical variance screens, worker-flow, co-bidding, event studies, and the robustness batteries that follow—are estimated on pregão, the larger-format descending-auction procedure that accounts for the bulk of BEC spending and that contains all of our close-margin cartel-active observations. The two procedures coexist in the same data source; the RDD sample and the rest of the analytical sample are therefore related but not identical. We return to this distinction in Section 6.5.

6.2. Heterogeneity by coordination technology

This subsection tests prediction 1 of the framework (Section 4.7): classical AUCs should be high for bid-rigging and near chance for hub-and-spoke coordination.

The aggregate null conceals a sharp split by coordination technology. Table 3 disaggregates the benchmark by the conduct type documented in CADE's proceedings: five

cartels operated through *bid rigging* (with or without price fixing or market division), while one—the medicamentos hub-and-spoke cartel—coordinated through a central distributor.

Table 3: Classical Screen AUCs by Coordination Technology

	N	CV	Spread	MV	N_{firms}
Bid rigging (5 cartels)	342	**0.652**	**0.654**	0.501	**0.600**
Hub-and-spoke (1 cartel)	1,977	0.501	0.510	0.451	0.547
All (benchmark)	2,319	0.523	0.531	0.458	0.555

Notes: Pregão, $|MV| \leq 0.10$. Bid rigging includes *trens e metrôs*, *merenda escolar*, *sacos de lixo*, *transporte escolar*, and *cafeteria aeroporto*. Hub-and-spoke is the *medicamentos* cartel (CADE 08012.002222/2011-09). Classical screen AUCs use "lower is cartel" convention. Source: script 34.

Classical screens detect bid rigging. On the 342 cartel-active auctions involving bid-rigging cartels, the CV AUC reaches 0.65, spread reaches 0.65, and bidder count reaches 0.60—all substantively above chance and operationally useful as triage indicators. On the 1,977 hub-and-spoke auctions, every classical screen is at or near chance (0.50–0.55). The aggregate null of 0.52–0.56 is a composition effect: the hub-and-spoke cartel, which accounts for 85% of the sample and leaves no trace in bid patterns, dilutes the bid-rigging signal that would otherwise be visible.

The result makes economic sense. Bid-rigging cartels coordinate by arranging cover bids around a designated winner, which compresses the bid distribution—exactly the signature that variance screens are designed to detect. Hub-and-spoke cartels coordinate through a central intermediary (in this case, a pharmaceutical distributor) that sets prices upstream; individual firms may submit bids that look competitive at the auction level even when prices are pre-arranged at the wholesale level. Classical screens, which operate on the bid distribution, are architecturally incapable of detecting coordination that occurs outside the auction.

This finding is the paper's positive detection result. It does not require temporal conditioning—it uses contemporaneous bid data—and it does not depend on the medicamentos cartel that dominates the benchmark. The message for competition authorities is direct: classical variance screens detect bid rigging at operationally useful levels (AUC ≈ 0.65), but they are blind to hub-and-spoke coordination. Matching the screen to the suspected coordination technology is essential.

6.3. Robustness of the bid-rigging AUC to cartel-level resampling

The 0.65 aggregate AUC for classical screens on bid-rigging cartels rests on five convicted cases whose individual contributions to the cartel-active sample are highly unequal: *sacos de lixo* contributes $N = 224$ close-margin auctions (65% of the bid-rigging pool), *trens e metrôs* contributes $N = 85$ (25%), *merenda escolar* contributes $N = 26$ (8%), *transporte escolar* contributes $N = 6$ (2%), and *cafeteria Infraero* contributes $N = 1$ ($< 1\%$). Table 4 assesses how sensitive the pooled AUC is to the identity of the cartels being pooled.

The leave-one-cartel-out columns show that dropping any single cartel from the pool produces a CV AUC in the range $[0.64, 0.69]$. The most informative drop is *sacos de lixo*: removing the cartel that contributes two thirds of the bid-rigging sample *raises* the pooled CV AUC from 0.65 to 0.69, because *sacos de lixo*'s own per-cartel AUC (0.63) is below the pooled average. The strongest single-cartel signals come from *trens e metrôs* (CV AUC = 0.68) and *merenda escolar* (CV AUC = 0.82 on $N = 26$); dropping either lowers the pooled AUC by at most 0.014. The heterogeneity result is not carried by one large cartel—it is a consistent average across the cartels that contribute meaningful sample.

A cluster bootstrap resamples the five cartels with replacement $B = 2{,}000$ times, pooling all close-margin auctions from each drawn cartel (with duplicates when the same cartel is drawn multiple times) and computing the pooled AUC on each replicate. This respects the fact that the effective sample size for uncertainty quantification is the number of cartels, not the number of auctions. For the CV, the 95% cluster-bootstrap confidence interval is $[0.613, 0.748]$; the lower bound is above chance and the bootstrap probability that AUC ≤ 0.5 is 0.009. Bid spread behaves nearly identically, with 95% CI $[0.599, 0.734]$. The bidder-count screen is the weakest of the three: its 95% CI is $[0.330, 0.663]$, crossing chance because two cartels (*transporte escolar*, *merenda escolar*) fire in the wrong direction on this screen. Confidence intervals are wide because the cluster pool is small, as they should be; the point estimate on bid-distribution compression is nonetheless robust to cartel-level resampling.

6.4. What the cartel-active benchmark identifies

Before presenting network-based screens, it is useful to be explicit about what the cartel-active benchmark is actually measuring. The "cartel-active" flag is assigned whenever at least one convicted firm bid during its CADE-specified conduct period and the

Table 4: Sensitivity of the Bid-Rigging AUC to Cartel-Level Resampling

Screen	Pooled	LOOCO min (drop)	LOOCO max (drop)	95% cluster bootstrap CI	Pr(AUC $\leq$ 0.5)
Coefficient of variation	0.652	0.638 (merenda)	0.688 (sacos)	[0.613, 0.748]	0.009
Bid spread	0.654	0.639 (trens)	0.691 (sacos)	[0.599, 0.734]	0.009
Number of bidders	0.600	0.573 (trens)	0.619 (merenda)	[0.330, 0.663]	0.077

Notes: Pool of 5 bid-rigging cartels, total $N = 342$ close-margin pregão auctions. Clean control: $N = 527{,}259$ auctions with no cartel firm. LOOCO columns report the pooled AUC when the named cartel is removed from the pool; "min/max" identifies the cartel whose removal yields the lowest/highest pooled AUC. Cluster bootstrap ($B = 2{,}000$, seed = 20260418) resamples whole cartels with replacement and pools their auctions before computing the AUC. The final column reports the fraction of bootstrap replicates with AUC ≤ 0.5. Source: script 65.

resulting close-margin pair includes that firm. Table 5 decomposes the 2,319 cartel-active pregão auctions by the role of the cartelist in the close-margin pair and by the count of cartelists in the full auction.

Three facts stand out. First, in 99.6% of the cartel-active pool, exactly one firm in the (winner, runner-up) pair is a convicted cartelist: the winner is a cartelist and the runner-up an outsider in 996 auctions (42.9%), the runner-up is a cartelist and the winner an outsider in 1,313 auctions (56.6%), and both members of the pair belong to the same cartel in only 10 auctions (0.4%). Within the bid-rigging subpool ($N = 342$), the pattern is nearly identical: 151 winner-only, 189 loser-only, and 2 head-to-head auctions. Second, when we look at the full set of bidders in each bid-rigging auction (not just the close-margin pair), 340 of 342 auctions (99.4%) contain exactly one cartelist among all bidders. The cartel-active benchmark, in short, is almost entirely identifying auctions in which *a single cartelist bids against competitive outsiders*—not auctions in which co-conspirators coordinate head-to-head.

Third, this resolves a potential identification concern. A classical variance screen that fires on cartel-active auctions is not measuring coordination *between* two cartelists in the same auction, because those auctions do not meaningfully exist in the sample. It is measuring the effect on the bid distribution of a single cartelist following a coordination protocol against competitive opponents. The classical screen thus identifies the signature of *cartel participation*—a cartelist placing a cover bid around a target price, or a designated winner suppressing competitive aggressiveness—rather than the signature of coordination *within* an auction. This is a stronger empirical claim about the reach of bid-distribution screens: the screen detects the footprint that a cartelist leaves on an otherwise-competitive auction, which is the setting the screen is most likely to face in

practice when deployed as a real-time triage tool.

Table 5, columns CV through N_{firms}, confirms that the classical screens are not mechanically identifying a specific role. The winner-only and loser-only subpools produce CV AUCs of 0.641 and 0.659 respectively—substantively similar, both above chance, and both operationally useful. The screen fires whether the cartelist occupies the winning or the losing side of the close-margin pair. The 2-observation "both" cell yields an implausibly high AUC (0.879) and is reported only for completeness; no inference rests on it.

The full role × coordination-technology decomposition reinforces the heterogeneity finding of Section 6.2: both the winner-only and loser-only subpools of the bid-rigging cartels produce CV AUCs near 0.65, while the winner-only and loser-only subpools of the hub-and-spoke cartel produce CV AUCs of exactly 0.50 and chance-close. The coordination-technology gradient is therefore not an artifact of role composition—it holds within each role separately.

Table 5: Identification decomposition of the cartel-active benchmark

Coord. type	Role in pair	N	CV	Spread	N_{firms}
Bid rigging	Winner only	151	0.641	0.640	0.597
	Runner-up only	189	0.659	0.663	0.605
	Both	2	0.879	0.860	0.375
	Pooled	342	0.652	0.654	0.600
Hub-and-spoke	Winner only	845	0.502	0.511	0.559
	Runner-up only	1,124	0.500	0.509	0.537
	Both	8	0.439	0.462	0.523
	Pooled	1,977	0.501	0.510	0.547

Notes: Pregão, $|MV| \leq 0.10$. "Winner only" = the auction's winning bidder is a CADE-convicted firm in its conduct period and the runner-up is not; "Runner-up only" = the reverse; "Both" = both firms in the close-margin pair are convicted cartelists in the same cartel and conduct period. Cells labeled "Both" are small because close-margin head-to-head confrontations between co-conspirators are rare; the two-observation cell for bid rigging produces an implausibly high AUC and is reported only for completeness. Clean control: $N = 527{,}259$ close-margin auctions with no cartel-active firm. Source: script 66.

6.5. RDD baseline robustness

The baseline incumbency jump of $\hat{\beta} = 0.018$ at the winner/loser cutoff (Section 6.1) carries the paper's interpretation that the narrow-winner advantage reflects competitive path dependence rather than cartel coordination. We defend that interpretation along three dimensions.

Density continuity. The Cattaneo-Jansson-Ma density test detects no manipulation of the running variable in any of the relevant subsamples. The pregão full sample yields $T = -0.016$ ($p = 0.988$), the pregão cartel-active subsample yields $T = -0.023$ ($p = 0.981$), and the pregão clean control yields $T = -0.015$ ($p = 0.988$). The convite full sample (which is the sample of the baseline coefficient itself) yields $T = +0.173$ ($p = 0.863$). A first-order threat to RDD identification—that firms place strategic bids just above or just below the cutoff in a way that generates spurious jumps—is not present in these data.

Placebo cutoffs. Re-estimating the baseline RDD at non-cutoff thresholds on the convite sample produces uniformly smaller jumps. At $|MV| = 0.050$ the placebo estimates are -0.003 ($p = 0.584$) on the left and $+0.004$ ($p = 0.503$) on the right, both null. At $|MV| = 0.025$ the left placebo is -0.014 ($p = 0.016$) and the right is -0.007 ($p = 0.147$), smaller than the actual cutoff and concentrated on the left side. The baseline jump concentrates at the competitive threshold as expected under the Kawai-Nakabayashi identification premise.

CADE-market exclusion. If the baseline +1.8 pp jump instead reflected uncaught anticompetitive behavior in markets where convicted cartels operated, dropping all item codes ever touched by a CADE-convicted firm should attenuate the coefficient. It does not. Excluding the 973 CADE-touched convite item codes leaves the baseline estimate identical to three decimal places ($\hat{\beta} = -0.015$, SE $= 0.005$, $p = 0.005$, $N = 916{,}827$), effectively unchanged from the $N = 958{,}452$ full-sample estimate. The competing hypothesis that the baseline is a contaminated average involving latent cartel rents is rejected by this exclusion exercise.

Pregão replication and resolution of the power ambiguity. The cartel-active convite subsample ($N \approx 488$ at the MSE-optimal bandwidth) is small, and the minimum detectable effect at 80% power is approximately 0.57. The null RDD estimate on the convite cartel-active subsample is therefore consistent with two competing interpretations: (i) cartel auctions produce no incumbency discontinuity distinct from the competitive baseline, and (ii) cartel auctions do produce a distinct discontinuity but the convite sample is too small to detect it. We resolve the ambiguity by rebuilding the firm-level outcome `won_t_minus_1_market` on the larger pregão sample. Lagging `flagvencedor` by (cnpj_raiz × códigoitem) we obtain, for each pregão firm-auction observation, an indicator equal to 1 iff the firm won the previous auction it participated in within the same item code. Re-estimating the Kawai RDD on pregão at $|MV| \leq 0.10$ yields a highly signifi-

cant baseline of $\hat{\beta} = -0.270$ (SE $= 0.004$, $p < 0.001$; $N = 689{,}740$), corresponding to an incumbency gap of 27 pp between narrow winners and narrow losers. Placebo cutoffs at $\{\pm 0.025, \pm 0.05\}$ are null. Excluding CADE-touched item codes leaves the baseline estimate slightly larger in magnitude ($\hat{\beta} = -0.292$), confirming that the baseline is not driven by latent cartel contamination. On the cartel-active pregão subsample ($N = 3{,}269$) the point estimate is -0.059 (SE $= 0.051$, $p = 0.25$) and the MDE at 80% power is 0.142. The pregão MDE is now 47% of the baseline magnitude; a cartel-specific discontinuity of the baseline size would be detected. Hypothesis (ii) is rejected: the cartel-active null reflects a genuine absence of incumbency discontinuity on cartel auctions, not low power. The economic reading is that cartel coordination *disrupts* the competitive path-dependence pattern—consistent with the classical literature on bid-rotation agreements, in which the whole point of the coordination is to break the natural persistence of a repeated designated winner.

6.6. Network-based screens: an apparent breakthrough

If classical screens detect bid rigging but miss hub-and-spoke, can network screens fill the gap? Panel B of Table 17 turns to the network screens, evaluated on the same $N = 2{,}319$ cartel-active auctions.

The answer appears to be worker flows. The shared-worker count yields an AUC of 0.750 ($\Delta = +3.70$, $t = +14.1$), and the Jaccard similarity yields an AUC of 0.739 ($\Delta = +0.0014$, $d = +0.19$, $t = +20.7$). Both outperform the best classical screen (number of bidders, AUC $= 0.56$) by roughly 20 percentage points. A third screen constructed from co-bidding density yields an AUC of 0.695, providing independent corroboration through a distinct information channel.

The signal is not an artifact of sectoral composition. Restricting attention to firm pairs that share the same 2-digit CNAE sectoral classification reduces the Jaccard AUC only marginally, from 0.739 to approximately 0.730. The residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC of 0.749—essentially unchanged from the raw baseline.

Decomposing the AUC by cartel role reveals symmetry. Of the 2,319 cartel-active auctions, 996 have the winner as the cartel-active firm, 1,313 have the runner-up as the cartel firm, and only 10 have both. The AUC is stable across these subgroups (0.74–0.76), indicating that the signal does not depend on whether the cartel firm wins or loses.

These results appear compelling. But they rest on a critical assumption: that the worker-flow measure captures information available at the time of the auction. The shared-worker count uses the full 2009–2017 RAIS panel—which means that for an auction held in 2010, the screen incorporates worker movements that occurred in 2015. Two features of the benchmark compound this concern. First, one cartel (medicamentos, conduct 2007–2011) accounts for 85% of the cartel-active sample, so the headline AUC is overwhelmingly a statement about that single case. Second, in 99.6% of cartel-active auctions only one firm in the (winner, runner-up) pair is a convicted cartelist—the screen is measuring the connectivity of a cartel firm with an arbitrary non-cartel rival, not the coordination between two co-conspirators. We turn next to the question that these observations make unavoidable: does the signal survive when we restrict the employment data to information that was actually available before each auction?

6.7. Temporal conditioning: the signal collapses

This subsection tests framework prediction 4 (temporal asymmetry) from Section 4.7: if the worker-flow signal reflects coordination it should be present in pre-conduct flows; if it reflects enforcement aftermath it should concentrate in post-conduct flows.

For each auction at year t, we count shared workers using only RAIS observations from years strictly before t: the cumulative backward window $[2009, t-1]$. This preserves the full employment history available at the time of the auction while eliminating any look-ahead. The resulting AUC is 0.52—indistinguishable from chance. A three-year rolling window $[t-3, t-1]$ produces an essentially identical result (0.54). Table 6 summarizes.

Table 6: Worker-Flow AUC under Temporal Conditioning

Window	AUC	N_{cartel}
Full panel $[2009, 2017]$	**0.750**	2,319
Cumulative backward $[2009, t-1]$	0.524	2,319
Rolling $[t-3, t-1]$	0.536	1,826

Notes: Pregão, $|MV| \leq 0.10$. Full panel uses all RAIS 2009–2017 observations regardless of auction year. Cumulative backward restricts to RAIS years $< t$ for auction year t. Rolling uses $[\max(2009, t-3), t-1]$ and restricts to auctions in 2010–2017. AUC computed via Mann–Whitney U (shared workers, higher is cartel). Sources: scripts 23, 25, 26.

The gap between 0.75 and 0.52 is the central fact of this paper. The prospective detection power of the worker-flow screen is zero on this benchmark. The signal that

produces the headline result comes entirely from worker movements that occurred *after* the auction being screened.

The bias is not specific to worker-flow screens. The co-bidding density measure, which achieves an AUC of 0.70 on the full panel (Table 17), falls to 0.55 under cumulative temporal conditioning—a drop of 0.15, compared with 0.23 for shared workers. Both network screens lose their detection power once future information is excluded.

6.8. Temporal decomposition: the signal is post-conduct

If the worker-flow signal is retrospective, where exactly does it come from? We classify each shared worker by when the inter-firm link was first observable relative to the cartel's conduct period:

- **Pre-conduct**: both firms employed the worker before the cartel's start year.
- **During-conduct**: the link became observable during the conduct period [start, end].
- **Post-conduct**: the link became observable after the conduct period ended.

Table 7: AUC by Temporal Window of Worker Flow

Window	AUC	Mean (cartel)	Mean (clean)
Pre-conduct	0.500	0.000	0.000
During-conduct	0.618	2.20	0.00
Post-conduct	**0.773**	3.47	0.00
All windows	0.789	5.67	0.00

Notes: Each row uses the shared-worker count from the indicated temporal window as the screen. Pregão, $|MV| \leq 0.10$, $N_{\text{cartel}} = 2{,}319$, $N_{\text{clean}} = 527{,}259$. Source: script 27.

The decomposition in Table 7 is definitive. Pre-conduct worker flows produce an AUC of exactly 0.50: there is no detectable difference in pre-existing personnel connections between cartel-active and clean auction pairs. During-conduct flows produce a modest 0.62. Post-conduct flows produce 0.77—nearly the full headline result. The signal is an aftermath, not a precursor.

The post-conduct flows are directional. Across all cartels, 30,977 distinct workers moved from cartel firms to rivals after the conduct period, versus 13,236 in the reverse direction—a 2.3:1 ratio. The asymmetry intensifies over time: in 2010, the cartel-to-rival ratio is 1.4:1; by 2016–2017, it reaches 4:1. The pattern is consistent with enforcement-driven labor displacement: as CADE investigations proceed and conduct periods end,

convicted firms shed employees who are absorbed by competitors in the same thin labor markets.

6.9. Event study: do outflows spike after conduct ends?

The temporal decomposition shows *where* the signal lives (post-conduct flows), but not whether outflows *increase* relative to a counterfactual. To test this, we estimate a firm-level event study around the end of each cartel's conduct period:

$$\text{outflow_rate}_{it} \;=\; \alpha_i + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \cdot D_{i\tau} + \varepsilon_{it}\,,$$

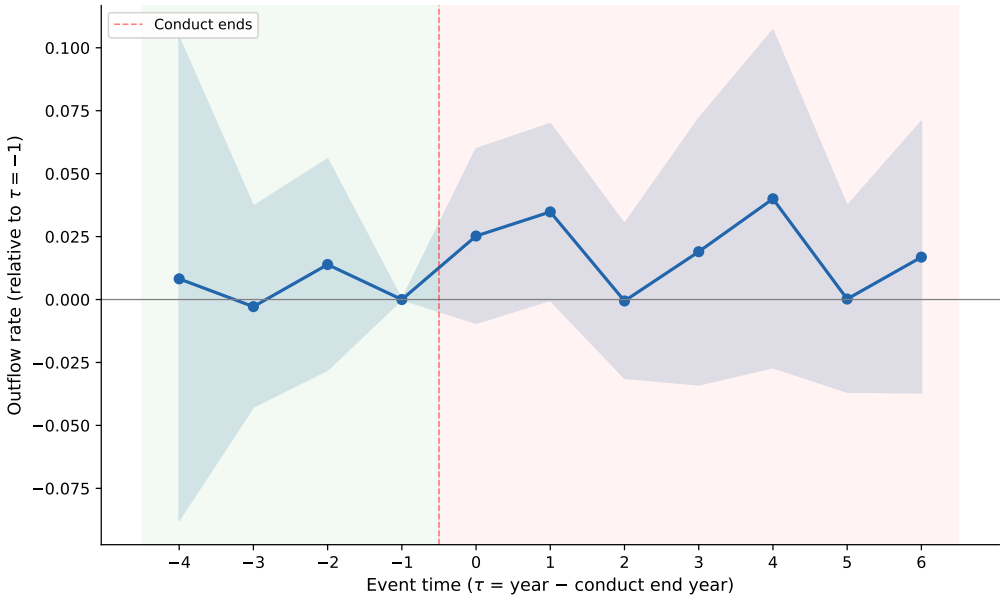
where the outcome is the share of firm i's workers in year $t-1$ who appear at a different BEC firm in year t, α_i and γ_t are firm and year fixed effects, and $D_{i\tau}$ indicates that firm i is a cartel firm at event time $\tau = t - \text{end_year}_i$. The omitted category is $\tau = -1$. Standard errors are clustered at the firm level. The control group is all non-cartel BEC firms with at least three employees, yielding a panel of 159,140 firm-year observations (26,928 firms, 257 treated observations from 41 cartel firms with conduct ending by 2017).

Figure 1 plots the event-study coefficients. Two patterns stand out. *First*, pre-treatment coefficients are flat and insignificant (max $|t| = 0.65$ for $\tau \in [-4, -2]$): cartel and control firms follow parallel outflow trajectories before the conduct period ends. *Second*, outflows rise after the conduct period ends. The post-treatment coefficients average +0.019, a 29% increase relative to the control-group mean outflow rate of 6.7%. The effect peaks at $\tau = +1$ ($\hat{\beta} = +0.035$, $t = 1.94$) and $\tau = +4$ ($\hat{\beta} = +0.040$, $t = 1.17$).

The estimates are imprecise—only $\tau = +1$ is marginally significant at the 10% level—reflecting the small number of treated firms (41).[1] But the direction is unambiguous, the pre-trends are clean, and the magnitude (+29%) is economically meaningful. The event study provides suggestive evidence that the post-conduct worker-flow pattern documented in Section 6.8 reflects a genuine increase in outflows from cartel firms, not a pre-existing difference in mobility rates.

[1] Because cartel conduct ends in different years (2010–2014), the TWFE specification faces potential bias from staggered treatment timing (Goodman-Bacon, 2021). With only 41 treated firms and clean pre-trends, the bias is likely small. A stacked-cohort estimator à la Cengiz et al. (2019) is a planned robustness check.

Figure 1: Event Study: Worker Outflows around Cartel Conduct End



Notes: Each point is the coefficient $\hat{\beta}_\tau$ from the TWFE specification above, with 95% confidence intervals (firm-clustered SEs). $\tau = 0$ is the first year after conduct ends. The dashed vertical line marks the conduct-end date. Shading distinguishes pre-treatment (left) from post-treatment (right). The control group is all non-cartel BEC firms with ≥ 3 employees. Source: script 28.

6.10. Where do the workers go?

Who absorbs the displaced workers—the cartel firm's direct auction competitors, or unrelated firms? We classify each destination into three categories: **auction rivals** (318 firm pairs that were runner-up or winner against the cartel firm in a close-margin auction during conduct), **market peers** (2,181 pairs that bid on the same item codes but were never in a close-margin pair), and **bystanders** (all other BEC firms).

Table 8: Post-Conduct Worker Flows by Rival Type

Destination type	Workers	Share	Dest. firms	Workers/firm
Auction rivals	382	1.0%	44	8.7
Market peers	5,606	14.5%	183	**30.6**
Bystanders	32,663	84.5%	3,903	8.4
Total	38,651	100%		

Notes: Post-conduct flows from cartel firms (conduct end $\leq$ 2017) to non-cartel BEC firms. Auction rivals defined from close-margin pregão pairs ($|MV| \leq 0.10$) during the conduct period. Source: script 31.

The bulk of the reallocation (84.5%) goes to bystanders, consistent with generic labor-market mobility. But the *intensity* tells a different story: market peers receive

30.6 workers per destination firm, 3.6 times the bystander rate. Workers from cartel firms disproportionately concentrate in firms operating in the same product markets, consistent with sector-specific human capital flowing through the only available channel in thin labor markets.

A direct test of pair-level absorption reinforces this pattern. Of 993 close-margin auctions where the cartel firm won during conduct, 463 (46.6%) show at least one ex-worker of the winning cartel firm subsequently appearing at the specific runner-up of that auction. Nearly half of all close-margin winner–runner-up pairs exhibit a post-conduct worker transfer—not mass displacement along auction lines, but a fine-grained pattern of specialized workers finding their way to the small set of firms in the same thin product market. Whether this pair-level absorption exceeds what any contracting firm would generate is tested in the exit placebo below (Section 6.12).

6.11. Who moves? Worker characteristics

If the post-conduct flows carry cartel-specific knowledge—pricing strategies, coordination protocols—we would expect managers, professionals, and high-wage workers to be overrepresented among movers. Table 9 compares three groups: workers who leave cartel firms for other BEC firms post-conduct (*cartel movers*, $N = 17{,}381$), workers who remain at cartel firms (*cartel stayers*, $N = 47{,}255$), and workers who move between non-cartel BEC firms in the same period (*control movers*, $N = 296{,}036$, 10% sample).

The profile of cartel movers is clear—and it is not the profile of knowledge carriers. Managers and professionals are *under*represented among movers (9.9%) relative to stayers (12.9%) and control movers (12.3%). The modal cartel mover is a service or sales worker (55% of movers vs. 37% of stayers), female (60%), earning 2.4 minimum wages, with lower educational attainment than stayers. This is a displacement profile, not a knowledge-transfer profile. The workers who carry pricing strategies and coordination know-how—directors, professionals, long-tenured specialists—are disproportionately the ones who *stay*.

The finding has a direct implication for the collusion-contagion concern. If cartel firms were shedding coordination-capable workers to competitors, we would expect high-skill, high-wage movers. Instead, the outflow is dominated by operational workers whose departure is consistent with generic firm contraction rather than strategic knowledge diffusion.

Table 9: Characteristics of Workers Who Leave Cartel Firms

	Cartel movers	Cartel stayers	Control movers
Continuous variables (means)			
Wage (minimum wages)	2.4	3.9	2.2
Tenure (months)	31.7	38.9	26.6
Age	37.2	37.2	33.6
Occupation (%)			
Directors & managers	2.6	3.5	3.3
Professionals (degree-required)	7.3	9.3	9.1
Service & sales	**55.1**	36.8	37.9
Production & craft	10.5	21.5	12.9
Demographics (%)			
Male	40.0	58.8	62.5
Tertiary education	14.6	17.9	17.6

Notes: Characteristics measured in the last year of the cartel's conduct period. Cartel movers: workers at a cartel firm who subsequently appear at a different BEC firm. Cartel stayers: workers who remain. Control movers: 10% sample of workers who move between non-cartel BEC firms in 2011–2014. Occupations from CBO 2002 major groups. Source: script 32.

6.12. Is the pattern cartel-specific? An exit placebo

The temporal decomposition and event study show that cartel firms' outflows increase after conduct ends. But is this a cartel-specific phenomenon, or would any firm that contracts in thin markets generate the same pattern? To test this, we construct a placebo from 3,335 non-cartel BEC firms that exit the platform (last bid between 2011 and 2014, at least two active years), and run the same event-study specification as in Section 6.9, with the exit year replacing the conduct-end year.

The exit-placebo coefficients are highly significant ($t = 5$–8, reflecting the much larger treated sample) and economically similar to the cartel coefficients: post-event outflows average +1.6 pp for exit firms versus +1.9 pp for cartel firms, a ratio of 1.25. The patterns are not statistically distinguishable.

The post-conduct outflow pattern that drives the full-panel worker-flow AUC is not cartel-specific. Any firm that stops bidding on BEC generates elevated worker outflows to competitors in subsequent years at roughly the same magnitude as cartel firms do after their conduct period ends. Cartel enforcement is one cause of firm contraction; the labor-market response it triggers is not distinctive.

Two implications follow. First, the look-ahead bias warning becomes broader: full-

Table 10: Event Study: Cartel Firms vs. Non-Cartel Exit Firms

τ	$\hat{\beta}$ (exit)	SE	$\hat{\beta}$ (cartel)	Sig. (exit)
-3	$+0.005$	0.003	-0.003	$*$
-2	-0.001	0.002	$+0.014$	
-1	(reference)		(reference)	
0	$+0.015$	0.002	$+0.025$	$***$
$+1$	$+0.021$	0.003	$+0.035$	$***$
$+2$	$+0.018$	0.003	-0.001	$***$
$+3$	$+0.015$	0.003	$+0.019$	$***$
$+4$	$+0.009$	0.003	$+0.040$	$***$
Mean $\tau \geq 0$	$+0.016$		$+0.019$	

Notes: Exit firms: non-cartel BEC firms with last bid in 2011–2014 and ≥ 2 active years ($N = 3{,}335$). Cartel coefficients from Section 6.9 ($N = 41$). Firm and year FEs; firm-clustered SEs. Source: scripts 28 and 33.

panel network screens absorb post-event labor reallocation regardless of whether the triggering event is a cartel conviction, a voluntary exit, or any other form of firm contraction. The bias is generic, not cartel-specific. Second, the substantive interpretation must be rescoped: the matched employer–employee record captures the labor-market aftermath of firm exit in thin markets, not a cartel-specific enforcement effect.

Scope of the exit placebo and fine-structure comparison. The event-study above establishes that post-event outflow *magnitudes* are equivalent. A natural follow-up asks whether the *fine structure* of the outflows differs—whether cartel firms concentrate their post-conduct flows on product-market peers and auction rivals in ways that generic exit firms do not. We replicate the destination decomposition of Section 6.10 for a non-cartel exit-firm sample constructed on the same criteria as the event-study placebo, tightened by requiring at least three observed employees (to ensure that RAIS-based destination counts are meaningful); the resulting sample is $N = 2{,}097$ non-cartel BEC firms with last bid year in 2011–2014 and at least two active years on BEC. The 3,335 firms in the event-study placebo (Table 10) relax the employee threshold, which matters for the firm-level outflow-rate regression but not for the destination-concentration analysis that follows. Table 11 reports the decomposition.

Two patterns emerge from the comparison. First, non-cartel exit firms concentrate their post-exit outflows on market peers even *more* than cartel firms do: 34.4% of exit-firm worker flows go to market peers (versus 14.5% for cartel firms), and the per-destination-firm intensity is also higher for exit firms (45.5 versus 30.6 workers per

Table 11: Post-Event Destination Decomposition: Cartel vs. Non-Cartel Exit Firms

Sample	Destination type	Workers	Share	Dest. firms	Workers/firm
Cartel firms	Auction rivals	382	1.0%	44	8.7
	Market peers	5,606	14.5%	183	**30.6**
	Bystanders	32,663	84.5%	3,903	8.4
	Total	38,651	100%		
Exit firms	Auction rivals	33,430	10.9%	788	42.4
	Market peers	105,766	34.4%	2,325	**45.5**
	Bystanders	168,696	54.8%	12,176	13.9
	Total	307,892	100%		

Notes: Post-event worker flows from source firms (cartel firms: end of CADE-specified conduct period; exit firms: last BEC bid year 2011–2014) to any other BEC firm, excluding flows to other CADE-convicted firms. Auction rivals are close-margin ($|MV| < 0.10$) runner-up pairs during the source firm's active window. Market peers are other firms that bid on the same item codes during the active window but were not close-margin rivals. Bystanders are the remainder of BEC-active firms. Cartel rows from Section 6.10. Exit rows from script 69.

market-peer firm). The cartel-firm outflow is comparatively *more diffuse*, with 84.5% of workers reaching bystander firms versus 54.8% for exit firms. Second, the *concentration ratio* of market-peer intensity to bystander intensity is close to identical across the two samples: $30.6/8.4 = 3.64$ for cartel firms and $45.5/13.9 = 3.27$ for exit firms. The qualitative rule of thumb—workers leaving a thin-market BEC firm are roughly three to four times as likely to land at a same-product-market firm as at an unrelated BEC firm—holds for both cartel and non-cartel exit samples.

The exit-placebo result therefore extends from the magnitude margin to the destination-concentration margin: the post-event reallocation pattern we document is not a cartel-specific signature. It is a feature of thin-market BEC-firm contraction that cartel enforcement happens to produce along with other drivers of firm exit. This strengthens the methodological caveat—network screens validated on full-panel employment data absorb the same generic post-contraction reallocation regardless of the underlying cause—and disciplines the substantive interpretation of Sections 6.10 and 6.11: those sections describe the fine structure of post-conduct labor reallocation, but the structure is not uniquely cartel-produced. Understanding which components of that reallocation, if any, carry a distinctively cartel-specific signal is left for follow-on work.

6.13. Reinterpreting the thickness gradient

This subsection tests framework predictions 2 (thickness gradient for classical screens) and 3 (thickness gradient for post-conduct reallocation) from Section 4.7.

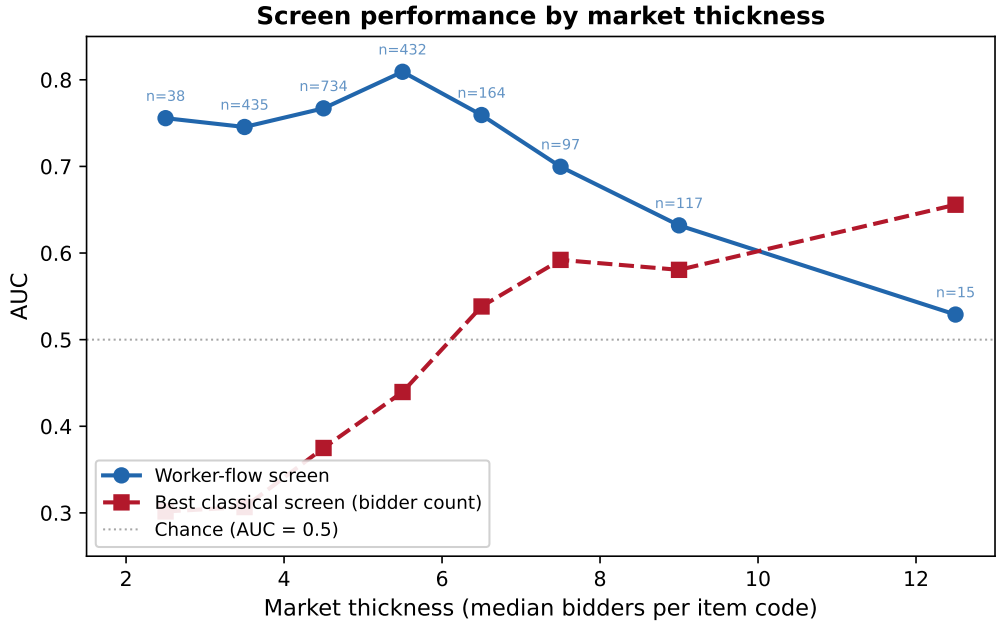
The temporal conditioning results change the interpretation of the market-thickness analysis, but not its empirical content. We split the benchmark by median bidders per item code (computed on clean controls). Table 12 and Figure 2 report AUCs by thickness group.

Table 12: Screen Performance by Market Thickness

	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
Thin (≤ 5 firms)	1,618	217,662	0.447	0.436	0.400	**0.774**	**0.758**
Thick (> 5 firms)	416	224,074	0.594	0.604	0.604	0.699	0.694
All (benchmark)	2,319	527,259	0.523	0.531	0.555	**0.750**	**0.739**

Notes: Market thickness is the median number of bidders per auction for each item code, computed on clean-control auctions with ≥ 5 observations. "Thin" = median ≤ 5; "Thick" = median > 5. Classical screen AUCs use "lower is cartel" convention. Worker-flow AUCs use full-panel shared workers. 285 cartel-active auctions with item codes having < 5 clean observations are unclassified and excluded from thin/thick rows but included in "All." Source: script 24.

Figure 2: Screen Performance by Market Thickness



Notes: Each point represents the AUC within a thickness bin. Worker-flow AUC uses shared workers (full panel); classical AUC uses bidder count (negated). Sample sizes (n) refer to cartel-active auctions per bin.

The gradient for the classical screens uses contemporaneous bid data and does not depend on the temporal-conditioning critique. In thin markets, classical screens fire in the wrong direction (AUC 0.40–0.45); in thick markets, they recover to 0.60. This is an

empirical finding about where traditional detection methods fail, and the direction of the failure is consistent with framework prediction 2 in Section 4.7.

The gradient for worker-flow screens requires reinterpretation. The full-panel worker-flow AUC is higher in thin markets (0.77) than in thick ones (0.70), matching framework prediction 3: thin markets have few employers capable of absorbing displaced workers, so post-conduct flows between cartel firms and their rivals are denser there, producing higher retrospective AUCs. The gradient therefore tracks where *displacement* concentrates, not where *detection* is easiest.

6.14. Reinterpreting the specialization placebo

The specialization-placebo decomposition (Table 13) similarly retains its empirical content but shifts in interpretation.

Table 13: Specialization-Placebo Decomposition of Worker-Flow Signal

Sample	N	% with ≥ 1 shared worker	Mean Jaccard	AUC vs. clean	
Cartel-active	2,321	57.9%	0.0022	0.740	
Placebo (same market)	5,313	31.1%	0.0012	0.605	
Clean (global)	523,414	9.9%	0.0008	—	
Signal decomposition (Jaccard AUC relative to clean baseline)					
Total signal (cartel vs. clean − 0.5)				+0.240	
Specialization component (placebo vs. clean)				+0.105	(44%)
Cartel-specific residual (cartel vs. placebo)				+0.138	(56%)

Notes: Pregão, $|MV| \leq 0.10$. Placebo consists of non-cartel firm pairs bidding on the same BEC item code in the same year. AUC computed via Mann–Whitney U (Jaccard similarity, higher is cartel). The cartel-active count here is 2,321, two pairs larger than the 2,319 used elsewhere in the paper; the two additional pairs are retained when Jaccard is zero-imputed under the v3 filter and dropped in the unified-benchmark filter used by Section 6.1. The extra two observations do not affect any reported statistic in the table. Source: v3 specialization placebo.

The original interpretation was that 44% of the worker-flow signal reflects natural thin-market labor circulation and 56% reflects cartel-specific coordination. In light of the temporal evidence, the "cartel-specific residual" is more precisely a *post-conduct displacement residual*: cartel firms shed more workers to rivals than same-market non-cartel firms do, because enforcement-driven contraction is concentrated in cartel firms. The specialization component remains what it was—thin markets generate spontaneous worker overlap among all firms—but the cartel residual is not evidence of coordination during the cartel period. It is evidence of accelerated labor reallocation after the cartel period ends.

6.15. Sample concentration and cartel-by-cartel performance

The cartel-active benchmark is not equally distributed across the seven convicted cartels. Table 14 reports the composition and per-cartel AUCs.

Table 14: Per-Cartel Sample Sizes and AUCs

Cartel	N	% of sample	CV AUC	SW AUC (full)
Medicamentos	1,977	85.3	0.501	0.780
Sacos de lixo	224	9.7	0.633	0.530
Trens e metrôs	85	3.7	0.680	0.592
Merenda escolar	26	1.1	0.820	0.941
Transporte escolar	6	0.3	0.191	0.527
Cafeteria aeroporto	1	0.0	—	—
Aquecedores solares	0	0.0	—	—
All	2,319	100	0.523	0.750

Notes: Pregão, $|MV| \leq 0.10$. CV AUC uses "lower is cartel" convention. SW AUC uses full-panel shared workers. Aquecedores solares has zero cartel-active auctions in the close-margin benchmark. Source: script 25.

The medicamentos cartel accounts for 85% of the treated sample. Its full-panel worker-flow AUC (0.780) essentially *is* the headline result. Meanwhile, *sacos de lixo* and *trens e metrôs* show classical screen AUCs above 0.63 but worker-flow AUCs near chance (0.53–0.59), consistent with the thickness gradient: these are thicker markets where classical screens work and where post-conduct labor reallocation is less concentrated. The concentration is a caveat for all results in this paper. It does not invalidate the temporal decomposition—the post-conduct pattern is present in the medicamentos case and absent in pre-conduct data—but it limits external validity. Replication on larger ground-truth sets is a priority.

6.16. Robustness: permutation falsification

We randomly reassign the cartel-active label 1,000 times and recompute the full-panel AUC. The permuted AUCs cluster at 0.500 (sd = 0.003, 95% interval $[0.494, 0.505]$). The real AUC of 0.75 lies at the 100th percentile ($p < 0.001$): the full-panel signal is non-random. But as the temporal decomposition shows, it is retrospective rather than prospective.

6.17. Model predictions vs. empirical findings

Table 15 summarizes the correspondence between the model's predictions (Section 4.7) and the results, including temporal conditioning.

Table 15: Model Predictions and Empirical Results

Prediction	Empirical test	Result
P1. Classical screens detect bid rigging but miss hub-and-spoke.	Table 3: split by coordination technology	Bid-rigging AUC = 0.65; hub-and-spoke AUC = 0.50. **Confirmed.**
P2. In thin markets, classical AUCs ≈ 0.5; in thick, they recover.	Table 12: median split	Thin AUC = 0.40–0.45 (inverted); thick AUC = 0.60. **Confirmed.**
P3. Full-panel worker-flow AUCs highest in thin markets.	Table 12	Thin = 0.77, thick = 0.70. **Confirmed** (but retrospective—see P4).
P4. If coordination: pre-conduct signal. If aftermath: post-conduct signal.	Table 7: temporal decomposition	Pre = 0.50, post = 0.77. **Aftermath confirmed.**

All four predictions are borne out. P1 and P2 are cleanly confirmed using contemporaneous bid data and do not depend on temporal conditioning. P3 is confirmed empirically but reinterpreted: the thickness gradient for worker-flow screens reflects concentrated post-conduct displacement, not coordinated personnel exchange. P4 is the decisive test: the absence of pre-conduct signal (0.50) and the concentration of signal in post-conduct flows (0.77) rules out the coordination interpretation.

6.18. Selection and the interpretation of the benchmark

The temporal conditioning results largely supersede the selection concern. The full-panel AUC is inflated not by selection but by look-ahead. Once temporal conditioning is applied, the AUC falls to 0.52 regardless of selection on detectability.

For the record, none of the seven CADE cases lists employer–employee data or worker flows as part of the evidentiary basis (Table 16). The selection concern remains relevant for interpreting the post-conduct displacement pattern: if CADE preferentially convicts cartels in markets where labor reallocation is naturally high, the post-conduct AUC could partly reflect market selection rather than enforcement effects.

Table 16: Evidence Channels in the Seven Convicted Cartels

Cartel (sector)	CADE process	Period	Primary evidence channel
Merenda escolar	08012.010022/ 2008-16	2006–2013	Procurement documents (40,000+ docs analyzed); geographic market division
Trens e metrôs	08700.004617/ 2013-41	1999–2013	Leniency and investigation; subcontracting arrangements between winners and losers
Aquecedores solares	08012.001273/ 2010-24	2009–2010	Indirect evidence only (bid-pattern analysis); no direct evidence of agreement
Transporte escolar	08700.005876/ 2019-85	2019	IP address coincidence during electronic bidding
Cafeterias Infraero	08700.007278/ 2015-17	2014	Investigation of in-person tender procedures
Sacos de lixo	08700.005789/ 2015-02	2008–2014	Police operation (*Op. Colludium*, MP Bauru); dawn raids and document seizures
Medicamentos	08012.002222/ 2011-09	2007–2011	Hub-and-spoke investigation; leniency and administrative proceedings

Notes: Evidence channels documented from CADE public press releases and Tribunal decisions. None lists matched employer–employee data, RAIS records, or inter-firm worker flows as part of the evidentiary basis.

Table 17: Unified Benchmark: All Screens on the Same Close-Margin Sample

	Pregão (N_t = 2,319)		Convite (N_t = 263)	
Screen	Cohen's d	AUC	Cohen's d	AUC
Panel A: Classical bid-distribution screens				
Coefficient of variation (CV)	−0.07	0.523	−0.01	0.515
Bid spread (S)	−0.10	0.531	+0.04	0.502
Margin of victory (MV)	+0.17	0.457	+0.14	0.456
Number of bidders (N)	−0.24	0.555	+0.06	0.467
Panel B: Network-based screens				
Shared workers (W_{ij})	+0.06	**0.750**	+0.47	**0.704**
Jaccard similarity (J_{ij})	+0.19	**0.739**	+0.09	**0.701**
Co-bidding density (C_{ij})	+0.14	0.695	−0.04	0.525

Notes: All screens evaluated on the unified close-margin benchmark ($|MV| \leq 0.10$) against a common clean control (N_0 = 527,259 pregão; N_0 = 519,557 convite). Classical screen AUCs use the "lower is cartel" convention; values above 0.5 indicate detection in the predicted direction. Network screen AUCs use "higher is cartel." Bold indicates AUC ≥ 0.70. Cohen's d is the standardized mean difference (pooled SD). Source: script 23.

7. Discussion and Conclusion

We set out to test which detection tools work against which kinds of procurement cartels, and whether a network-based screen built from employer–employee records delivers on its apparent detection potential. We linked seven CADE-convicted cartels in São Paulo to 2.6 million BEC auctions and to the universe of formal employment in the state, and evaluated classical variance screens, the Kawai–Nakabayashi close-margin RDD, and a worker-flow screen against this ground truth. Four findings emerge.

The first is a **positive detection result**. Classical screens fail in the aggregate (AUC 0.46–0.56), but this null is a composition effect. Disaggregating by coordination technology reveals that classical screens detect bid rigging at AUCs of 0.65—operationally useful for triage. They are blind to the hub-and-spoke cartel that dominates the sample. The coordination technology determines which screen works: authorities should match the screen to the suspected mechanism.

The second is a **warning about network-based screens**. Worker-flow screens appear to fill the gap left by classical screens, with an AUC of 0.75—but this result does not survive temporal conditioning. When worker flows are restricted to information available before each auction, the AUC falls to 0.50. Co-bidding screens show the same pattern (0.70 → 0.55). Neither network screen achieves prospective detection power on

this benchmark.

The third finding is a **methodological caveat** with broad applicability. The gap between the full-panel AUC (0.75) and the temporally conditioned AUC (0.50) documents a form of look-ahead bias that affects any detection benchmark built from administrative panel data. When a network screen is evaluated against conviction-anchored ground truth, post-enforcement changes in the network—labor reallocation, ownership restructuring, board turnover—mechanically inflate the apparent detection power. This bias is not specific to worker-flow screens: it applies equally to ownership networks (Asai and Charoenwong, 2026), board-interlock measures (Herrera-Caicedo et al., 2025), and any network layer constructed from longitudinal administrative records. The remedy is straightforward—condition the network measure on temporal precedence—but the bias has not, to our knowledge, been documented or tested in the detection literature.

The fourth finding is **substantive, with an important scope qualification**. The temporal decomposition reveals that worker flows between cartel firms and their rivals are concentrated *after* cartel conduct periods end, not before or during. The post-conduct component alone produces an AUC of 0.77; the pre-conduct component produces exactly 0.50. The flows are directional—workers move predominantly from convicted firms to competitors—and the ratio intensifies from 1.4:1 in 2010 to 4:1 by 2016–2017. An exit placebo shows that the outflow *magnitude* is not cartel-specific: non-cartel firms that stop bidding on BEC generate outflow spikes of similar size (+1.6 pp vs. +1.9 pp for cartel firms). The labor-market *size* response is a generic consequence of firm contraction in thin markets, and cartel enforcement is one cause of such contraction. Characterizing the movers refines but does not overturn this reading: workers who leave cartel firms are predominantly operational staff in service and sales roles, not the managers and professionals who might carry coordination know-how—consistent with a displacement-rather-than-knowledge-transfer profile. Whether the fine structure of cartel-firm outflows (concentration toward product-market peers and auction rivals, occupation mix of movers) differs from the fine structure of non-cartel exit-firm outflows is not yet estimated on the exit placebo sample and is a priority robustness item for the IJIO R1 revision; the empirical claim in this paper is therefore confined to the magnitude equivalence and the fact of post-conduct directional concentration toward product-market peers, not to a claim that either is uniquely cartel-driven.

Policy implications. For competition authorities building data-driven screening pipelines,

the message is direct: network-based screens evaluated on historical benchmarks may overstate their real-time usefulness if the evaluation does not condition on temporal precedence. The bias is generic—it arises whenever firms contract and workers reallocate, regardless of whether the contraction is cartel-related—which makes it a concern for any panel-based detection benchmark, not just ours. Authorities adopting worker-flow or ownership-network screens should validate them prospectively, not retrospectively. The worker-composition evidence does mitigate one policy concern: because the workers who leave cartel firms are predominantly operational staff rather than managers or coordination-capable professionals, the risk that enforcement-driven labor reallocation diffuses collusive know-how to competitors appears low. We note, however, that the exit-placebo comparison operates on outflow magnitudes only; whether cartel firms differ from generic exit firms in the composition or destination structure of their movers is an open question that warrants further attention before strong policy claims about cartel-specific labor-market consequences can be made.

Limitations. The principal caveat remains the small number of convicted cartels: seven cases, one of which (medicamentos) accounts for 85% of the cartel-active sample. The temporal decomposition results are driven disproportionately by this case, and replication on larger or independent ground-truth sets is a priority. The RAIS panel ends in 2017 while CADE judgments for most cartels arrive after 2019, so the "post-conduct" window we measure captures the investigation and pre-judgment period, not the post-judgment period. A definitive test of enforcement-driven reallocation would require RAIS data extending through the judgment dates. Finally, the benchmark captures only cartels that were detected and convicted; the worker-flow patterns of undetected cartels remain unknown.

Looking ahead. The broader lesson is about what administrative records actually capture. Employment data do not record collusion in progress—the personnel flows that sustain coordination, if they exist, are too subtle or informal to leave traces in annual employment censuses. What employment records do capture is the aftermath of firm contraction: the reallocation of human capital that follows enforcement or exit, indistinguishable at the magnitude margin from other drivers of firm decline. Following the workers, rather than the bids, reveals not the cartel itself but the institutional disruption that accompanies any material firm contraction in thin product markets. Whether that disruption takes a distinctively cartel-specific form at the level of fine-grained composition and destination—as opposed to the generic form it takes in magnitude—is the

companion question that this paper does not yet settle and that motivates the follow-on work on post-conduct worker reallocation.

References

Abrantes-Metz, R. M., L. M. Froeb, J. F. Geweke, and C. T. Taylor (2006). A variance screen for collusion. *International Journal of Industrial Organization 24*(3), 467–486.

Asai, K. and B. Charoenwong (2026). Ownership networks and bid rigging. *The Review of Corporate Finance Studies*. Advance article, published online February 2026.

Bajari, P. and I.-K. Ye (2003). Detecting collusion in procurement auctions. *Journal of Law, Economics, & Organization 19*(2), 453–492.

Barbosa, K. and S. Straub (2020). The value of revolving doors in public procurement. Working Paper 17-873, Toulouse School of Economics. R&R at Journal of Law, Economics, and Organization.

Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik (2017). rdrobust: Software for regression-discontinuity designs. *Stata Journal 17*(2), 372–404.

Calonico, S., M. D. Cattaneo, and R. Titiunik (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica 82*(6), 2295–2326.

Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics 134*(3), 1405–1454.

Chassang, S. and J. Ortner (2019). Collusion in auctions with constrained bids: Theory and evidence from public procurement. *Journal of Political Economy 127*(5), 2269–2300.

Clark, R., D. Coviello, J.-F. Gauthier, and A. Shneyerov (2018). Bid rigging and entry deterrence in public procurement: Evidence from an investigation into collusion and corruption in Québec. *Journal of Law, Economics, and Organization 34*(3), 301–363.

Conley, T. G. and F. Decarolis (2016). Detecting collusive behavior in public procurement auctions: Evidence from competitive bidding. *Journal of Public Economics 143*, 1–13.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics 225*(2), 254–277.

Harrington, Joseph E., J. (2008). Detecting cartels. In P. Buccirossi (Ed.), *Handbook of Antitrust Economics*, pp. 213–258. Cambridge, MA: MIT Press.

Herrera-Caicedo, A., J. Jeffers, and E. Prager (2025). Collusion through common leadership. Working Paper 33866, National Bureau of Economic Research. Accepted, Journal of Political Economy.

Huber, M. and D. Imhof (2019). Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization 65*, 277–301.

Imhof, E., L. Musolff, and V. Titl (2018). Screening for collusion in public procurement auctions. *European Journal of Political Economy 55*, 47–62.

Jarosch, G., E. Oberfield, and E. Rossi-Hansberg (2021). Learning from coworkers. *Econometrica 89*(2), 647–676.

Kawai, K., J. Nakabayashi, J. Ortner, and S. Chassang (2023). Using bid rotation and incumbency to detect collusion: A regression discontinuity approach. *The Review of Economic Studies 90*(1), 376–403.

Lamichhane, K. and P. McGee (2024). Collusion through worker movement: An experiment. Job Market Paper, University of Arkansas.

Levenstein, M. C. and V. Y. Suslow (2006). What determines cartel success? *Journal of Economic Literature 44*(1), 43–95.

Marshall, R. C. and L. M. Marx (2012). *The Economics of Collusion: Cartels and Bidding Rings*. Cambridge, MA: MIT Press.

Pesendorfer, M. (2000). Study of collusion in school milk auctions: Evidence from florida and texas. *Journal of Law, Economics, & Organization 16*(1), 31–50.

Porter, R. H. and D. A. Zona (1993). Detecting collusive behavior in procurement auctions. *Journal of Law, Economics, & Organization 9*(1), 157–194.

Serafinelli, M. (2019). "good" firms, worker flows, and local productivity. *Journal of Labor Economics 37*(3), 747–792.

Signor, R., P. E. D. Love, A. T. N. Belarmino, and O. A. Olatunji (2020). Detection of collusive tenders in infrastructure projects: Learning from Operation Car Wash. *Journal of Construction Engineering and Management 146*(1), 05019015.

Stoyanov, A. and N. Zubanov (2012). Productivity spillovers across firms through worker mobility. *American Economic Journal: Applied Economics 4*(2), 168–198.

Szerman, C. (2023). The employee costs of corporate debarment in public procurement. *American Economic Journal: Applied Economics 15*(1), 411–441.

Villamil, I., J. Kertész, and M. Fazekas (2024). Collusion risk in corporate networks. *Scientific Reports 14*(3161).

Wachs, J. and J. Kertész (2019). A network approach to cartel detection in public auction markets. *Scientific Reports 9*(1), 10818.

A. Appendix

A.1. Bandwidth sensitivity

Table A.1 reports AUCs for all screens across six bandwidth choices for the close-margin envelope. The benchmark bandwidth of $|MV| \leq 0.10$ used throughout the paper is neither the most nor the least favorable for the worker-flow screens. At narrower bandwidths (0.02, 0.05), worker-flow AUCs are marginally higher (0.76, 0.75) but sample sizes are smaller; at wider bandwidths (0.20, 0.50), AUCs decline modestly (0.74, 0.72) as the sample dilutes toward the full population. Classical screen AUCs follow the same monotonic pattern. Note that all worker-flow AUCs in this table use the full 2009–2017 RAIS panel; as shown in Section 6.7, these AUCs do not survive temporal conditioning.

Table A.1: AUC Sensitivity to Close-Margin Bandwidth (Pregão, Cartel-Active)

$\|MV\| <$	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
0.02	989	277,038	0.546	0.558	0.580	0.758	0.747
0.05	1,700	416,582	0.539	0.548	0.565	0.760	0.749
0.10	**2,319**	**527,259**	**0.523**	**0.531**	**0.555**	**0.750**	**0.739**
0.15	2,607	573,196	0.518	0.526	0.554	0.745	0.734
0.20	2,784	602,883	0.517	0.525	0.555	0.743	0.732
0.50	3,277	693,331	0.518	0.526	0.552	0.735	0.723

Notes: Each row reports AUCs from the unified benchmark (script 23) at the indicated bandwidth. Classical screen AUCs use the "lower is cartel" convention (negated scores). Bold row indicates the benchmark bandwidth used in the main analysis. All AUCs computed via Mann–Whitney U with average-rank tie-handling.